

XML GUIDE FOR DUALSPHYSICS

VARIABLE RESOLUTION METHOD



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<http://dual.sphysics.org>

Basis of the method

1. The Variable resolution approach available in DualSPHysics v5.4 is based on a domain decomposition method.
2. The domain is divided into an arbitrary number of different zones, which are connected only through additional Dirichlet boundary conditions enforced by buffer regions.
3. The physical quantities of SPH particles in the buffers are computed by means of a corrected SPH interpolation over adjacent sub-domains.
4. To model the mass transfer between the sub-domains, a procedure based on the evaluation of the Eulerian mass flux at the domain boundaries is applied. Particles that belong to a specific zone are created/destroyed in the buffer regions and do not interact with fluid particles that belong to a different resolution zone.

For additional information about the variable resolution formulation, please refer to:

- **F. Ricci, R. Vacondio, and A. Tafuni.** Multiscale smoothed particle hydrodynamics based on a domain-decomposition strategy. *Computer Methods in Applied Mechanics and Engineering*, 418:116500, 2024.

Command Line Arguments: PARAMETERS

-vres Enable Vres execution.

-vres_order:[0,1,2] Maximum order of accuracy for the interpolation of buffer particles:

- **0:** 0th order (Shepard interpolation)
- **1:** 1st order
- **3:** 2nd order.

-vres_method:[0,1] Interpolation algorithm:

- **0:** Liu-Liu correction algorithm (used also in mDBC and Inlet/Outlet)
- **1:** Moving-Least Square (MLS) algorithm .

-vr_threshold:[float] Threshold for the reciprocal of the scaled condition number.

-vres_fast:[0,1] Single precision calculation for buffer interpolation (GPU execution):

- **0:** Double precision
- **1:** Single precision.

Command Line Arguments: DEFAULT PARAMETERS

Default parameters:

- **-vres_order:1** 1st Order
- **-vres:method:0** Liu-Liu Correction
- **-vres_threshold:100**
- **-vres_fast:1** Single precision

Guidelines and Tips:

In case of instabilities at the interface:

- Lower the order of accuracy
- Decrease the threshold (a value <500 is suggested)
- Switch to MLS in the presence of free-surface (MLS more stable but less accurate than Liu-Liu correction)

XML file: **CaseDef-vres**: Definition of variable resolution zones

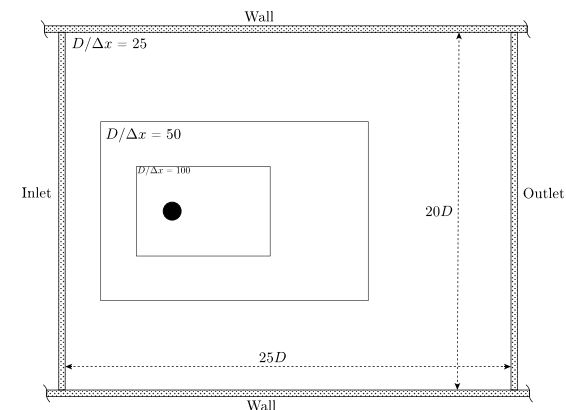
- To enable Vres, the `<vres></vres>` section must be added to the xml `<casedef></casedef>` configuration.
- The `<vres></vres>` section is structured to reflect the hierarchical structures of nested regions.
- The definition of a variable resolution zone is done in a `<bufferbox></bufferbox>` section.
- To create a refinement region inside a previously defined refinement region, a `<bufferbox></bufferbox>` section must be inserted inside the previously defined `<bufferbox></bufferbox>` section.
- It is possible to add the parameter `active` to `<vres></vres>` and `<bufferbox></bufferbox>` definition to activate and deactivate refinement regions.

Example: CaseFlowCylinder_Re200

```
<vres active="true">
  <buffsizeh v="2" comment="Buffer size according to H (H*v) (default v=2)" />
  <bufferbox>
    <!-- Refinement level 1 -->
    <point x="-0.30" y="0" z="-0.51" />
    <size x="1.5" y="0.0" z="1" />
    <dpratio v="2" comment="Refinement ratio." />
    <buffsizeh v="2" comment="Buffer size according to H (H*v) (default v=2)" />
    <simulationdomain useparent="true" comment="Defines domain of simulation"/>
    <bufferbox active="true">
      <!-- Refinement level 2 -->
      <point x="-0.10" y="0" z="-0.250" />
      <size x="0.75" y="0.0" z="0.5" />
      <dpratio v="2.0" comment="Refinement ratio." />
      <simulationdomain useparent="true" comment="Defines domain of simulation"/>
    </bufferbox>
  </bufferbox>
</vres>
```

Zone 2
 $D/dx=50$

Zone 3:
 $D/dx=100$



XML file: **CaseDef-vres-bufferbox: PARAMETERS AVAILABLE**

- **point:** starting point of a variable resolution zone.
- **size/endpoint:** define the extension of the variable resolution zone.
- **dpratio** refinement ratio between parent and child domain. A value $0.5 < v < 2.5$ is recommended
- **buffsizeh** Buffer size according to smoothing length h . The default value is $v=2$
- **overlappngh** Extension of the overlapping region according to length h . Only applied when **overlappngh** is explicitly declared in the Vres domain definition. Recommended when interface between Vres regions crosses solid boundaries and/or in presence of free-surface flows.
- **tracking** If enabled, the Vres domain moves according the translational motion of the moving or floating body specified by the **mkbound** value.
- **simulationdomain** define refinement region domain limits.

XML file: CaseDef-vres-bufferbox: POINTREF

When an interface between two resolution zones crosses a solid boundary, it is suggested to change the <pointref> value in the <definition></definition> section as:

<pointref x="#Dp/2" y="#Dp/2" z="#Dp/2" />

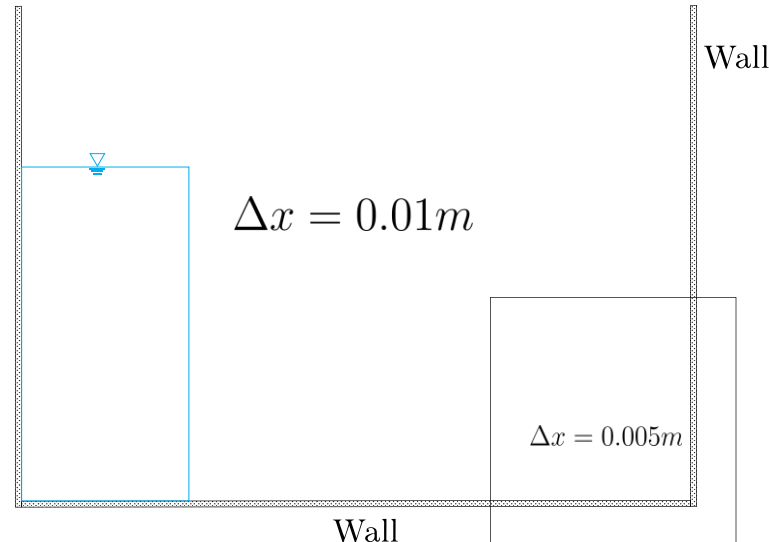
This allows for the correct definition of solid boundaries with different resolutions.

Example: DamBreak2D

Geometry definition

```
<definition dp="0.01" units_comment="metres (m)">
  <pointref x="#Dp/2" y="#Dp/2" z="#Dp/2" />
  <pointmin x="-0.1" y="0" z="-0.4" />
  <pointmax x="4.2" y="0" z="3.5" />
</definition>
<commands>
  <list name="GeometryForNormals">
    <setactive drawpoints="0" drawshapes="1" />
    <setshapemode>actual | bound</setshapemode>
    <setnormalinvert invert="true" />
    <!-- Tank -->
    <setmkbound mk="0" />
    <drawbox>
      <boxfill>bottom | left | right</boxfill>
      <point x="0" y="-1" z="0" />
      <size x="#size_x" y="2" z="#size_z" />
    </drawbox>
    <shapeout file="hdp" />
    <resetdraw />
  </list>
  <mainlist>
    <!-- Actual geometry at dp/2 -->
    <runlist name="GeometryForNormals" />
    <!-- Particles -->
    <setdrawmode mode="full" />
    <!-- Fluid -->
    <setmkfluid mk="1" />
    <drawbox>
      <boxfill>solid</boxfill>
      <point x="0" y="-1" z="0" />
      <size x="1" y="2" z="2" />
    </drawbox>
    <!-- Tank -->
    <setmkbound mk="0" />
    <drawbox>
      <boxfill>bottom | left | right</boxfill>
      <point x="0" y="-1" z="0" />
      <size x="#size_x" y="2" z="#size_z" />
      <layers vdp="0.5,1.5,2.5,3.5,4.5" />
    </drawbox>
  </mainlist>
</commands>
```

Wall



XML file: CaseDef-vres-bufferbox: SIMULATIONDOMAIN

- Limits of the refinement region must be explicitly defined in the **<bufferbox></bufferbox>** definition.
- When **<simulationdomain useparent="true"/>** the refinement region uses the domain limits of the parent region
- When multiple nested region are tracking an object, it is suggested to explicitly define the domain limits for each refinement region according to the expected movement of the object.

Example: FallingWedge2D

```
<vres >
  <bufferbox>
    <point x="-0.49" y="0" z="1.31" />
    <size x="0.98" y="0" z="0.59" />
    <dpratio v="2.0" comment="Refinement ratio" />
    <overlappingh v="2" comment="Overlapping size according to H (H*v) (default v=0)" />
    <tracking mkbound="51" comment="Reference to moving or floating object inside refinement region." />
    <simulationdomain comment="Defines domain of simulation (default=Uses minimum and maximum position of the generated particles)">
      <posmin x="default - 400%" y="default" z="default - 400%" comment="e.g.: x=0.5, y=default-1, z=default-10%" />
      <posmax x="default+100%" y="default" z="default+100%" />
    </simulationdomain>
  </bufferbox>
  <bufferbox>
    <point x="-0.39" y="0" z="1.40" />
    <size x="0.79" y="0" z="0.398" />
    <dpratio v="2.0" comment="Refinement ratio" />
    <overlappingh v="2" comment="Overlapping size according to H (H*v) (default v=0)" />
    <tracking mkbound="51" comment="Reference to moving or floating object inside refinement region." />
    <simulationdomain comment="Defines domain of simulation (default=Uses minimum and maximum position of the generated particles)">
      <posmin x="default - 400%" y="default" z="default - 400%" comment="e.g.: x=0.5, y=default-1, z=default-10%" />
      <posmax x="default+100%" y="default" z="default+100%" />
    </simulationdomain>
  </bufferbox>
</vres>
```

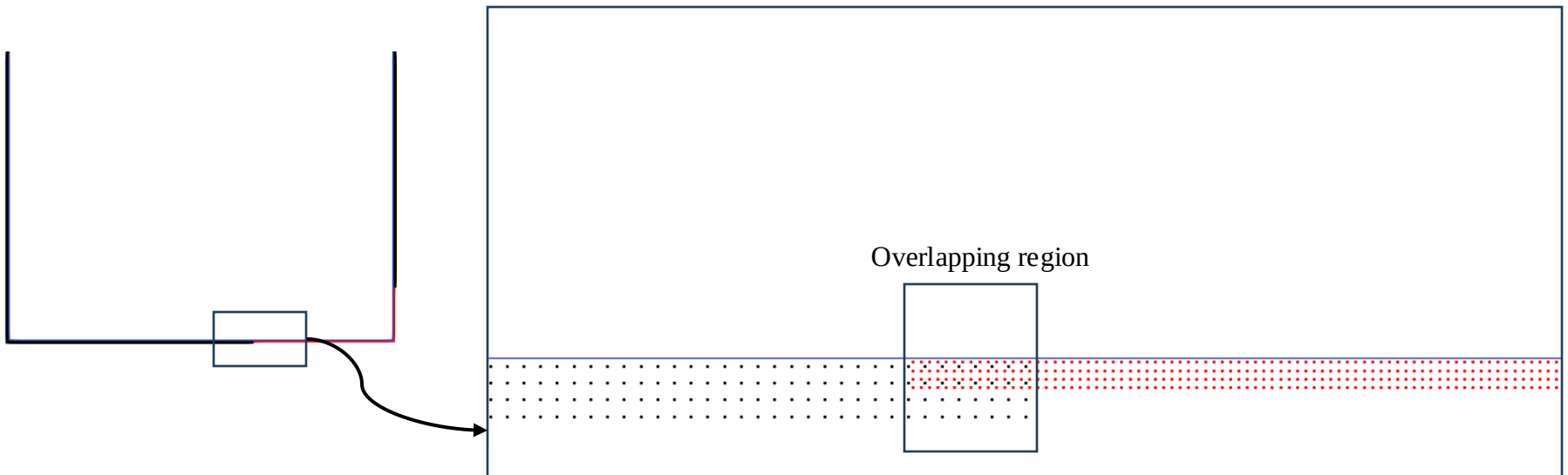

XML file: CaseDef-vres-bufferbox: OVERLAPPINGH

When an interface between two resolution zones crosses a solid boundary, or a free-surface, it is suggested to activate an overlapping region.

Overlapping region can be activated with **overlappingh**, that defines the extension of the overlapping region according to smoothing length h .

Example: DamBreak2D

```
<vres active="true">
  <buffsizeh v="2" comment="Buffer size according to H (H*v) (default v=2)" />
  <bufferbox>
    <point x="2.5" y="0" z="-0.20" />
    <endpoint x="4.1" y="0" z="0.60" />
    <dpratio v="2" comment="Refinement ratio." />
    <buffsizeh v="2" comment="Buffer size according to H (H*v) (default v=2)" />
    <overlappingh v="2" comment="Overlapping size according to H (H*v) (default v=0)" />
  </bufferbox>
</vres>
```



XML file: CaseDef-vres-bufferbox: TRACKING

- The variable resolution algorithm is capable of tracking objects, so that position of the object relative to the refinement region remain fixed.
- To enable the tracking, `<tracking></tracking>` section must be added to the `<bufferbox></bufferbox>` definition
- The object to be tracked is defined by **mkbound** value.

Example: FallingWedge2D

Wedge Geometry definition:

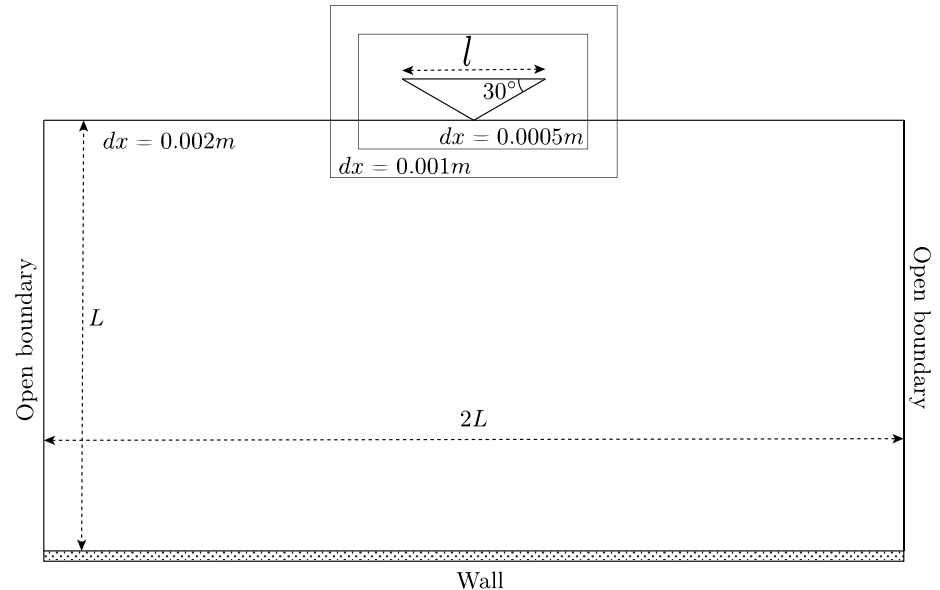
```
<setfrdrawmode auto="true" />
<setmkbound mk="51" />
<drawextrude closed="true">
  <extrude x="0" y="1" z="0" />
  <point x="0" y="-0.5" z="1.5" />
  <point x="0.25" y="-0.5" z="1.64433" />
  <point x="-0.25" y="-0.5" z="1.64433" />
  <layers vdp="0,1,2,3" />
</drawextrude>
<setfrdrawmode auto="false" />
```

Floating object definition:

```
<floatings>
  <floating mkbound="51">
    <massbody value="241" />
    <linearvelini x="0" y="0" z="-6.15" />
  </floating>
</floatings>
```

VRes Configuration:

```
<vres>
  <bufferbox>
    <point x="-0.49" y="0" z="1.31" />
    <size x="0.98" y="0" z="0.59" />
    <dratio v="2.0" comment="Refinement ratio" />
    <overlappinh v="2" comment="Overlapping size according to H (H*v) (default v=0)" />
    <tracking mkbound="51" comment="Reference to moving or floating object inside refinement region." />
    <simulationdomain comment="Defines domain of simulation (default=Uses minimum and maximum position of the generated particles)">
      <posmin x="default - 400%" y="default" z="default - 400%" comment="e.g.: x=0.5, y=default-1, z=default-10%" />
      <posmax x="default+100%" y="default" z="default+100%" />
    </simulationdomain>
  </bufferbox>
  <bufferbox>
    <point x="-0.39" y="0" z="1.40" />
    <size x="0.79" y="0" z="0.398" />
    <dratio v="2.0" comment="Refinement ratio" />
    <overlappinh v="2" comment="Overlapping size according to H (H*v) (default v=0)" />
    <tracking mkbound="51" comment="Reference to moving or floating object inside refinement region." />
    <simulationdomain comment="Defines domain of simulation (default=Uses minimum and maximum position of the generated particles)">
      <posmin x="default - 400%" y="default" z="default - 400%" comment="e.g.: x=0.5, y=default-1, z=default-10%" />
      <posmax x="default+100%" y="default" z="default+100%" />
    </simulationdomain>
  </bufferbox>
</vres>
```



Two refinement region that track the object with **mkbound=51**

XML file: Execution-special: SPECIAL CONFIGURATIONS

- Special configuration (InOut/Chrono/Flexstruct/etc..) must be explicitly activated for each refinement regions.
- Is possible to activate special configuration for a specific refinement region using **#VResId** variable in the Xml configuration.
- **#VResId==0** identify the outermost region.

Examples are available in the following folders:

- 02_CaseFlowCylinder_Re
- 05_FloatingDuck
- 06_Special

Example: CaseFlowCylinder_Re200

```
<inout active="#VResId=0">
  <memoryresize size0="2" size="4" comment="Initial memory resize (size0) and the following memory resizes according to initial inlet/outlet particles (default=2 and 4)" />
  <determinlimit value="1e-3" comment="Use 1e-3 for first_order or 1e+3 for zeroth_order (default=1e+3)" />
  <extrapolatemode value="3" />
  <!-- INFLOW BUFFER -->
  <inoutzone>
    <refilling value="1" comment="Refilling mode. 0:Simple full, 1:Simple below zsurf, 2:Advanced for reverse flows (very slow) (default=1)" />
    <inputtreatment value="2" comment="Treatment of fluid entering the zone. 0:No changes, 1:Convert fluid (necessary for outlet), 2:Remove fluid (recommended for inlet)" />
    <layers value="4" comment="Number of inlet/outlet particle layers" />
    <zone2d comment="Input zone for 2D simulations">
      <particles mkfluid="2" direction="right" />
    </zone2d>
    <imposevelocity mode="0" comment="Imposed velocity 0:fixed value, 1:variable value, 2:Extrapolated value (default=0)" />
    <velocity v="#fluidvel" comment="Uniform velocity" units_comment="m/s" />
    </imposevelocity>
    <imposerhop mode="0" comment="Outlet rhop 0:Imposed fixed value, 1:Hydrostatic, 2:Extrapolated from ghost nodes (default=0)" />
  </inoutzone>
  <!-- OUTFLOW BUFFER -->
  <inoutzone>
    <refilling value="1" comment="Refilling mode. 0:Simple full, 1:Simple below zsurf, 2:Advanced for reverse flows (very slow) (default=1)" />
    <inputtreatment value="1" comment="Treatment of fluid entering the zone. 0:No changes, 1:Convert fluid (necessary for outlet), 2:Remove fluid (recommended for inlet)" />
    <layers value="4" comment="Number of inlet/outlet particle layers" />
    <zone2d comment="Input zone for 2-D simulations">
      <particles mkfluid="3" direction="left" />
    </zone2d>
    <imposevelocity mode="2" comment="Imposed velocity 0:fixed value, 1:variable value, 2:Extrapolated value (default=0)" />
    <imposerhop mode="0" comment="Outlet rhop 0:Imposed fixed value, 1:Hydrostatic, 2:Extrapolated from ghost nodes (default=0)" />
  </inoutzone>
  <!-- END BUFFERS -->
</inout>
```

XML file: Execution-parameter: PERIODIC BOUNDARIES

- With Variable resolution, only the outermost region can have periodic boundaries
- To enable Periodic BC only for outermost domain, use **#VResId** variable

Example: CaseFlowCylinder_Re200

```
<parameter key="XPeriodicIncZ" value="0" comment="Increase of Z with periodic BC" active="#VResId==0" />
<simulationdomain comment="Defines domain of simulation (default=Uses minimum and maximum position of the generated particles)">
  <posmin x="default" y="default" z="default" comment="e.g.: x=0.5, y=default-1, z=default-10%" />
  <posmax x="default" y="default" z="default" />
</simulationdomain>
</parameters>
```

XML file: CaseDef-geometry: FILLBOX

- Special treatment is needed in case of variable resolution execution when a fillbox is used in the case definition
- If the seed point of the fillbox is not inside a refinement region, fluid particles will not be created. In case of refinement regions nested inside each other, it is suggested to define the seed point inside the innermost region (see 05_FloatingDuck).
- In case of disjoint refinement regions, it is suggested, keeping the same mkvalue, to defined multiple fillbox, each with a different seed point.

EXAMPLE:

Fillbox definition:

```
<setmkfluid mk="1"/>
  <fillbox x="3.0" y="0" z="0.26">
    <modefill>void</modefill>
    <point x="-1" y="-1" z="-1"/>
    <size x="20" y="2" z="#1.4+Dp/2"/>
  </fillbox>
  <fillbox x="5.0" y="0" z="0.26">
    <modefill>void</modefill>
    <point x="-1" y="-1" z="-1"/>
    <size x="20" y="2" z="#1.4+Dp/2"/>
  </fillbox>
```

VRes Configuration:

```
<vres active="true">
  <buffsizeh comment="Buffer size according to H (H*v) (default v=2)" v="2"/>
  <bufferbox active="true">
    <point x="3.0" y="-0.18" z="0.26" />
    <size x="0.5" y="0.36" z="0.44" />
    <dpratio v="2.0" comment="Refinement ratio" />
    <buffsizeh v="2" comment="Buffer size according to H (H*v) (default v=2)" />
    <overlappinh v="2"/>
    <tracking mkbound="20" comment="Reference to moving or floating object inside refinement region." />
    <simulationdomain>
      <posmin x="default-400%" y="default-100%" z="default - 400%" />
      <posmax x="default+100%" y="default+100%" z="default+100%" />
    </simulationdomain>
  </bufferbox>
  <bufferbox active="true">
    <point x="5.0" y="-0.18" z="0.26" />
    <size x="0.5" y="0.36" z="0.44" />
    <dpratio v="2.0" comment="Refinement ratio" />
    <buffsizeh v="2" comment="Buffer size according to H (H*v) (default v=2)" />
    <overlappinh v="2"/>
    <simulationdomain useparent="true" />
  </bufferbox>
</vres>
```